Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

Monish.P

Roll No. 250701446

EXPERIMENT 14

Understand feature scaling

Experiment 14: Understand feature scaling

Aim

To apply feature scaling to the given data.

Source Code

```
import numpy as np
import pandas as pd
df=pd.read_csv('/content/pre-process_datasample.csv')
df
df.head()
df.Country.fillna(df.Country.mode()[0],inplace=True)
features=df.iloc[:,:-1].values
df.Country.fillna(df.Country.mode()[0],inplace=True)
label=df.iloc[:,-1].values
from sklearn.impute import SimpleImputer
age=SimpleImputer(strategy="mean",missing_values=np.nan)
Salary=SimpleImputer(strategy="mean",missing_values=np.nan)
age.fit(features[:,[1]])
SimpleImputer()
Salary.fit(features[:,[2]])
SimpleImputer()
SimpleImputer()
SimpleImputer()
features[:,[1]]=age.transform(features[:,[1]])
features[:,[2]]=Salary.transform(features[:,[2]])
features
from sklearn.preprocessing import OneHotEncoder
oh = OneHotEncoder(sparse_output=False)
Country=oh.fit_transform(features[:,[0]])
Country
final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)
final_set
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)
feat_standard_scaler
from sklearn.preprocessing import MinMaxScaler
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)
feat_minmax_scaler
```

Output - Experiment 14

Country Age Salary Purchased
0 France 44.0 72000.0 No
1 Spain 27.0 48000.0 Yes
2 Germany 30.0 54000.0 No
3 Spain 38.0 61000.0 No
4 Germany 40.0 NaN Yes
5 France 35.0 58000.0 Yes
6 Spain NaN 52000.0 No
7 France 48.0 79000.0 Yes
8 NaN 50.0 83000.0 No
9 France 37.0 67000.0 Yes
Next steps: Generate code with df View recommended plots New interactive sheet
Country Age Salary Purchased
0 France 44.0 72000.0 No
1 Spain 27.0 48000.0 Yes
2 Germany 30.0 54000.0 No
3 Spain 38.0 61000.0 No
4 Germany 400 NaN Yes
Next steps: Generate code with df View recommended plots New interactive sheet
 SimpleImputer i ?
 SimpleImputer i ?
 SimpleImputer i ?
array([['France', 44.0, 72000.0],
['Spain', 27.0, 48000.0],
['Germany', 30.0, 54000.0],
['Spain', 38.0, 61000.0],
['Germany', 40.0, 63777.77777777778],
['France', 35.0, 58000.0],
['Spain', 38.77777777777778, 52000.0],
['France', 48.0, 79000.0],
['France', 50.0, 83000.0],
['France', 37.0, 67000.0]], dtype=object)
array([[1., 0., 0.],
[0., 0., 1.],
[0., 1., 0.],
[0., 0., 1.],
[0., 1., 0.],
[1., 0., 0.],
[0., 0., 1.],
[1., 0., 0.],
[1., 0., 0.],
[1., 0., 0.]])
array([[1.0, 0.0, 0.0, 44.0, 72000.0],
[0.0, 0.0, 1.0, 27.0, 48000.0],
[0.0, 1.0, 0.0, 30.0, 54000.0],
[0.0, 0.0, 1.0, 38.0, 61000.0],
[0.0, 1.0, 0.0, 40.0, 63777.77777777778],
[1.0, 0.0, 0.0, 35.0, 58000.0],
[0.0, 0.0, 1.0, 38.77777777777778, 52000.0],
[1.0, 0.0, 0.0, 48.0, 79000.0],
[1.0, 0.0, 0.0, 50.0, 83000.0],

Output - Experiment 14 (continued)

```
[1.0, 0.0, 0.0, 37.0, 67000.0]], dtype=object)
array([[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
7.58874362e-01, 7.49473254e-01],
[-1.00000000e+00, -5.00000000e-01, 1.52752523e+00,
-1.71150388e+00, -1.43817841e+00],
[-1.00000000e+00, 2.00000000e+00, -6.54653671e-01,
-1.27555478e+00, -8.91265492e-01],
[-1.00000000e+00, -5.00000000e-01, 1.52752523e+00,
-1.13023841e-01, -2.53200424e-01],
[-1.00000000e+00, 2.00000000e+00, -6.54653671e-01,
1.77608893e-01, 6.63219199e-16],
[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
-5.48972942e-01, -5.26656882e-01],
[-1.00000000e+00, -5.00000000e-01, 1.52752523e+00,
0.00000000e+00, -1.07356980e+00],
[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
1.34013983e+00, 1.38753832e+00],
[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
1.63077256e+00, 1.75214693e+00],
[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
-2.58340208e-01, 2.93712492e-01]])
array([[1. , 0. , 0. , 0.73913043, 0.68571429],
[0. , 0. , 1. , 0. , 0. ],
[0. , 1. , 0. , 0.13043478, 0.17142857],
[0. , 0. , 1. , 0.47826087, 0.37142857],
[0. , 1. , 0. , 0.56521739, 0.45079365],
[1. , 0. , 0. , 0.34782609, 0.28571429],
[0. , 0. , 1. , 0.51207729, 0.11428571],
[1. , 0. , 0. , 0.91304348, 0.88571429],
[1. , 0. , 0. , 1. , 1. ],
[1. , 0. , 0. , 0.43478261, 0.54285714]])
```

Result - Experiment 14

Feature scaling was demonstrated on the given data.